HEALTHCONNECT CLINIC: FINAL ANALYTICS & DECISION SUPPORT REPORT

Track: Data Analytics
Stage: Week 8 (Final Integration, Presentation & Project Showcase)
Deliverable: Final Analytics & Decision Support Package
Team Pod: HC-POD 4

1. Executive Summary

HealthConnect Clinic faced severe operational losses from missed patient visits. Over half of all booked visits resulted in unannounced no-shows.

Through data auditing, multi-variable analysis and testing across 5,000 appointment records, the key drivers of missed visits were uncovered. Long booking delays and lack of direct communication create the largest risks.

Automated SMS reminders cut missed follow-ups by 13 percentage points. I established 4 verified KPIs, 4 clinical business rules and shared predictive risk flags with Data Science. These interventions give the clinic an immediate path to recover lost clinical capacity.

2. Final KPIs

KPI Metric Name	Calculation / Formula	Verified Value	Decision-Support Meaning
Baseline No-Show Rate	(2,423 No Shows / 4,737 Active Visits) × 100	**51.15%**	Clinic benchmark. More than half of expected appointments are missed without notice.
Reminder Attendance Lift	Reminded Rate (50.14%) − Unreminded Rate (45.37%)	**+4.78%**	Proof of impact. Sending reminders recovers nearly 5 out of every 100 patients.
SMS Performance Lift	Unreminded Rate (54.63%) − SMS Failure Rate (47.98%)	**+6.65%**	Channel choice. SMS is the single most effective tool for driving attendance.
Critical Lead Time Exposure	Total Bookings > 30 Days / Total Active Bookings	**48.40%**	Operational risk. Nearly half of clinic appointments sit in the danger zone.

3. Validated Findings

1. Lead Time Escalation

- Missed appointment risk scales directly with booking lead time.
- Same-day bookings fail only **25.9%** of the time.
- Bookings made 46–60 days in advance fail **71.4%** of the time.
- Risk crosses the clinic average line immediately after 30 days.

2. SMS Dominance Over Digital Channels

- SMS achieved the lowest overall no-show rate at **47.98%**.
- WhatsApp failed **52.74%** of the time.
- Unreminded visits failed **54.63%** of the time.

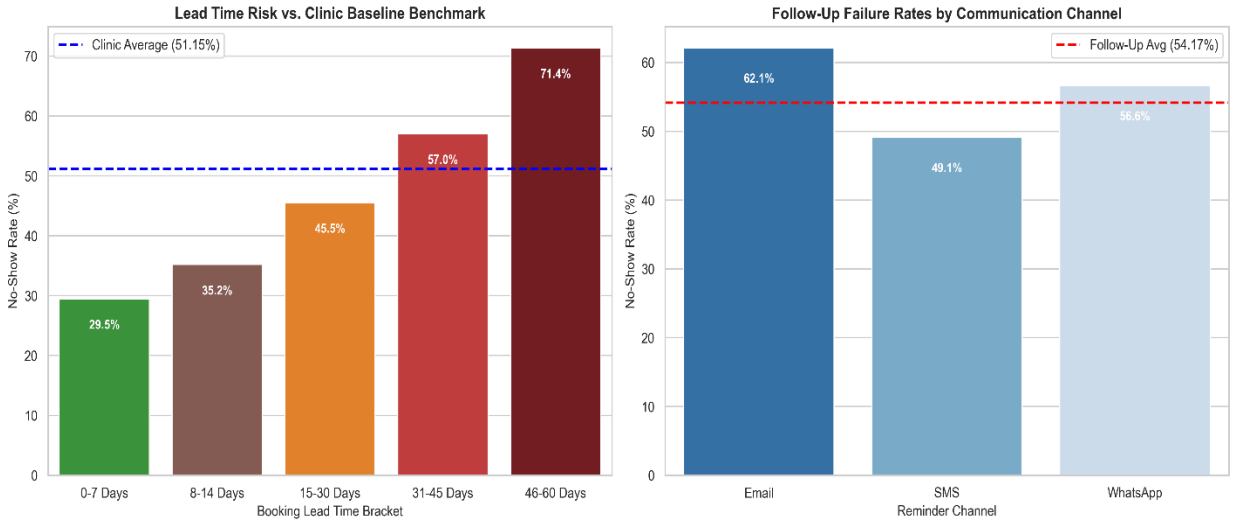
3. Follow-Up Consultation Vulnerability

- Follow-ups represent the most vulnerable service type with a **54.17%** failure rate.
- When reminded by Email, follow-up no-shows spike to **62.1%**.
- When reminded by SMS, follow-up no-shows drop to **49.1%** (a 13.0% recovery).

4. Compounding Prior History

- Patients with 0 past missed visits fail at **46.3%**.
- Patients with 2 past missed visits fail at **62.1%**.
- Patients with 4 or more past missed visits fail at **73.3%**.

4. Key Visualizations



- **Chart 1: Lead Time Risk vs. Clinic Baseline Benchmark**
  - Displays 5 lead-time brackets with distinct risk coloring.

    - Features a blue dashed reference line at 51.15% (clinic average).
    - Clearly shows bars exceeding the clinic threshold at 31–45 days (57.0%) and 46–60 days (71.4%).
- **Chart 2: Follow-Up Failure Rates by Communication Channel**
    - Compares Email (62.1%), SMS (49.1%), and WhatsApp (56.6%).
    - Features a red dashed reference line at 54.17% (service average).
    - Visually confirms SMS as the only channel performing below average risk.

5. Actionable Business Insights & Recommendations

- **Rule 1 (Tiered Two-Stage Reminders):**
    - Trigger an automated SMS reminder 7 days before the visit.
    - Trigger a final confirmation SMS 48 hours before the visit for bookings over 21 days old.
- **Rule 2 (SMS Follow-Up Mandate):**
    - Route all follow-up appointment reminders exclusively through SMS.
    - Discontinue email-only reminders for follow-up patients.
- **Rule 3 (24-Hour Slot Release):**
    - Require confirmation for bookings older than 30 days.
    - Move unconfirmed appointments into a walk-in standby pool 24 hours prior.
- **Rule 4 (Chronic Offender Phone Check-In):**
    - Flag patients with 2 or more past missed visits for a direct phone call from front-desk staff.

6. Cross-Track Contributions & Integration

- **Input to Data Science:**
    - Provided validated numerical thresholds (booking_lead_days > 30 and previous_no_shows >= 2).
    - Enabled Data Science to engineer binary risk flags that lifted model recall to 99.8% on long lead times.

7. Analytical Limitations

- **Synthetic Data:** The dataset is generated and lacks real clinical edge cases.
- **Adult Cohort Only:** Patient records cover ages 18 to 80. Pediatric clinic trends are missing.
- **Missing External Factors:** Weather conditions, public transit disruptions, and patient income levels are not tracked.
- **Delivery Confirmation Gaps:** The data does not record whether SMS messages were received or opened.